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EXPERIMENT 7

Managing Users, Roles and Permissions in Zoho Creator

1. Aim

To learn and implement the process of creating users, defining roles, and setting permissions in Zoho Creator, and to understand how access control can be managed effectively for different users in a business application.

2. Introduction

Zoho Creator provides a robust user management system that enables administrators to control access to applications. It uses a role-based access control (RBAC) model where users are assigned to specific roles, and permissions are given to control what actions they can perform on different modules and data. Roles help in organizing users based on their responsibilities, while permissions define the exact level of access such as view, create, edit, delete, and share. Proper user and permission management ensures data security, streamlines collaboration, and maintains the integrity of the application.

3. Objectives of the Study

- To understand the concept of users, roles, and permissions in Zoho Creator.
- To learn how to add and manage users in an application.
- To understand the role hierarchy and its impact on data access.
- To create and configure custom permissions for different roles.
- To explore data sharing options between peers and subordinates.
- To develop a complete user access control system for a real-world use case.

4. Procedure

- Log in to the Zoho Creator platform.
- Create a new application or open an existing application.
- Go to the Settings section and select Users & Permissions.
- Add users to the application by providing their email address and name.
- Allows roles based on the organizational hierarchy using the Roles tab.
- Assign appropriate roles to the users.
- Configure role details such as role name, parent role, description, and reports to.
- Create custom permission profiles using the Permissions tab.
- Set module-level permissions (Access, View, Create, Edit, Delete) for each profile.
- Configure data sharing settings (With peers, with subordinates, with parent role).
- Set API and security permissions if required.
- Test the configuration with different user accounts.
- Publish the application after completing the setup.

5. Illustrative Example

Consider a Food Delivery System in which different users such as CEO, Manager, Delivery Staff, and Customer interact with the application. Each user requires different levels of access to the application's data and functionality.

5.1 Implementation using Zoho Creator

Create a Food Delivery System application and configure users, roles, and permissions as follows:

Roles Hierarchy:

- CEO (Top-level role)
    - Manager (reports to CEO)
    - Delivery Staff (reports to Manager)
    - Customer (reports to Delivery Staff)

Permissions Profiles:

- CEO Profile – Full access to all modules
- Manager Profile – Create, View, Edit access
- Staff Profile – View and Edit access
- Customer Profile – View access to limited modules

User Assignment (Example):

- User: gokul.s2024@rajalakshmi.edu.in
- Name: GOKUL S
- Role: Delivery Staff
- Permission Profile: Staff Profile

5.2 Illustrative Design

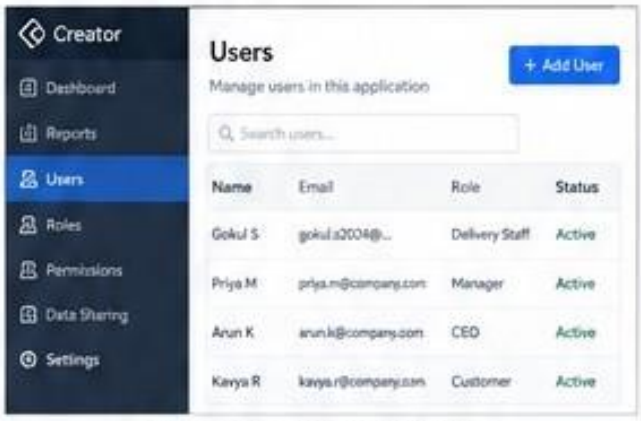


Fig 1. Users List in Zoho Creator

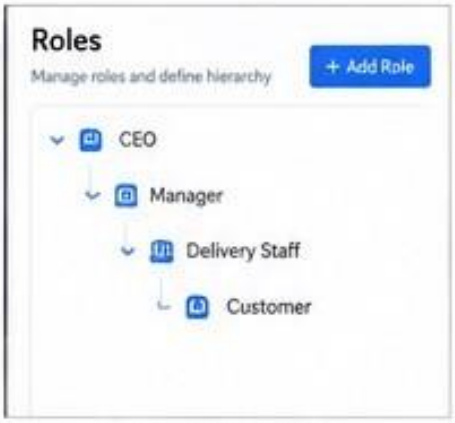


Fig 2. Roles Hierarchy

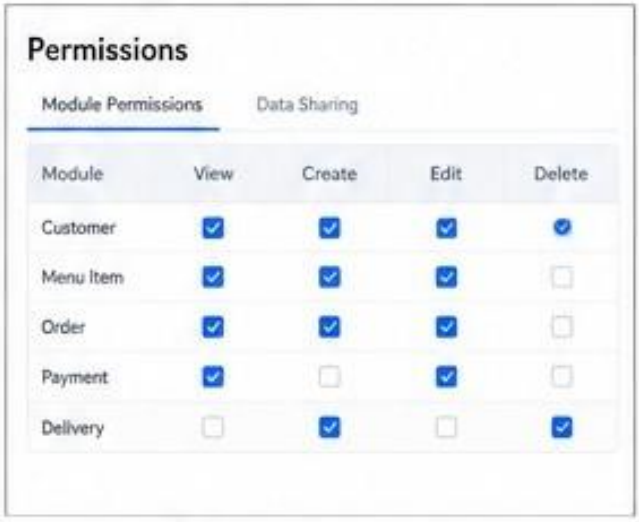


Fig 3. Module Permission Settings

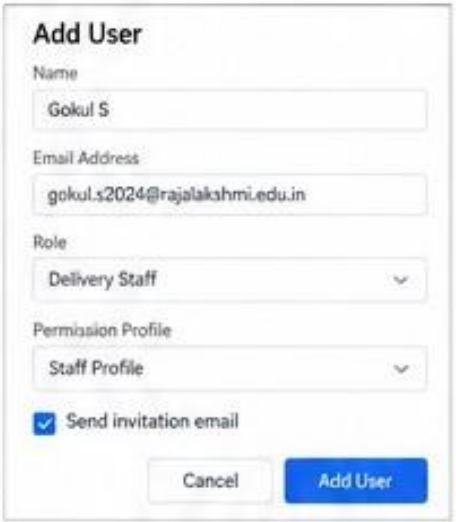


Fig 4. Add User Form

6. Advantages and Limitations

6.1 Advantages

- Provides a structured role-based access control system.
- Ensures data security by restricting unauthorized access.
- Simplifies user management through role hierarchy.
- Permissions can be configured at both user and role levels.
- Allows controlled data sharing between peers, subordinates, and parent roles.
- Reduces administrative effort by grouping users under roles.
- Helps maintain data integrity and accountability.

6.2 Limitations

- Complex permission structures may require careful planning.
- Incorrect configuration can lead to data leakage or access issues.
- Large organizations may need additional customization.
- Changes in role hierarchy may require reassigning users.
- Advanced permission requirements may need Deluge scripting.
- Managing a large number of users and roles can become difficult.

7. Result

The experiment was completed successfully by creating users, defining roles, and configuring permissions in Zoho Creator. The exercise demonstrated how users, roles, and permissions can be combined to implement a secure and well-structured access control system for a business application with minimal coding effort.